

Task 5: Controlled Generation via Classifier-Free Guidance

1. Introduction

Maintaining high semantic quality in low-data regimes is a persistent challenge for discrete diffusion. This task implements Classifier-Free Guidance (CFG) to steer the Encoder-Decoder model by interpolating between standard conditional logits and gradients from a secondary quality predictor.

2. Implementation Details

A 3-layer Quality Classifier (139.14K parameters) is trained to estimate target BERTScore from intermediate hidden states. During inference, we modify the predicted logits by a guidance scale λ : $z' = z + \lambda \cdot \nabla_h Q(h)$.

👉 **Implementation:** [analysis/quality_classifier.py](#)

3. Experimental Observations

The guidance sweep demonstrated that for the Encoder-Decoder variant, the optimal guidance scale is $\lambda = 0.0^{**}$. Increasing guidance beyond 1.0 introduced semantic "hallucinations"—where the model generated valid but incorrect Sanskrit tokens due to noisy gradients from the small classifier dataset (N<100).

3.1 Subtask Results Table (Optimal Model - T4)

Following the 6-point CFG analysis framework, the outcomes for the Encoder-Decoder T4 guidance sweep are detailed below:

Subtask Identification	Metric / Status	T4 Model Result
1. Predictor Implementation	3-Layer MLP Architecture Build	✅ Build: 139.14K parameters successful.
2. Hidden State Extraction	Latency impact of gradient hooks	~ 18.4% Penalty per step
3. Training Loss Convergence	Classifier MSE loss at epoch 50	0.0122 (Stable convergence)

Subtask Identification	Metric / Status	T4 Model Result
4. <i>Guidance Scale Sweep</i>	Optimal λ determination (BERT-F1 vs Diversity)	$\lambda = 0.0$ (Optimal stability limit)
5. <i>Hallucination Threshold</i>	Identification of λ -collapse point	$\lambda > 1.5$ (Syntax Breakdown)
6. <i>Final Production Strategy</i>	Conclusion for T4-deployment	Disable Guidance (Default state)

4. Analysis & Conclusion

While the CFG framework is mathematically integrated, its effectiveness is bottlenecked by the classifier's training data volume. For the Encoder-Decoder architecture, the T4 denoising loop provides sufficient baseline quality, and additional steering currently introduces more noise than refinement. Future scaling of the quality-labeled corpus remains a prerequisite for enabling non-zero guidance.